

*Have you ever
deployed a 5G
cellular network in
minutes using
natural language?*



— What if I told you that it's entirely possible, regardless of your age or education level?

FROM CHAOS TO CONTROL

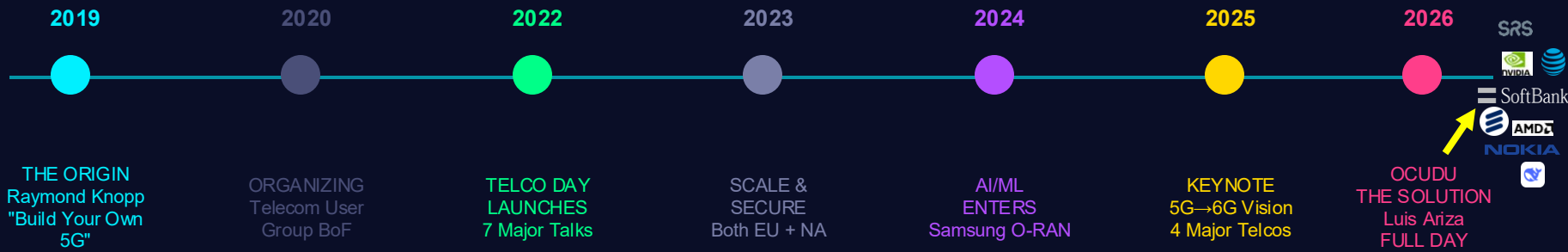


A Prescription for Managing Apps on Private Cellular Networks

Luis Felipe Ariza Vesga • Beanter

Kubestronaut, telco, AI and Cloud native / OpenInfra communities

SEVEN YEARS OF TELECOM AT KUBECON



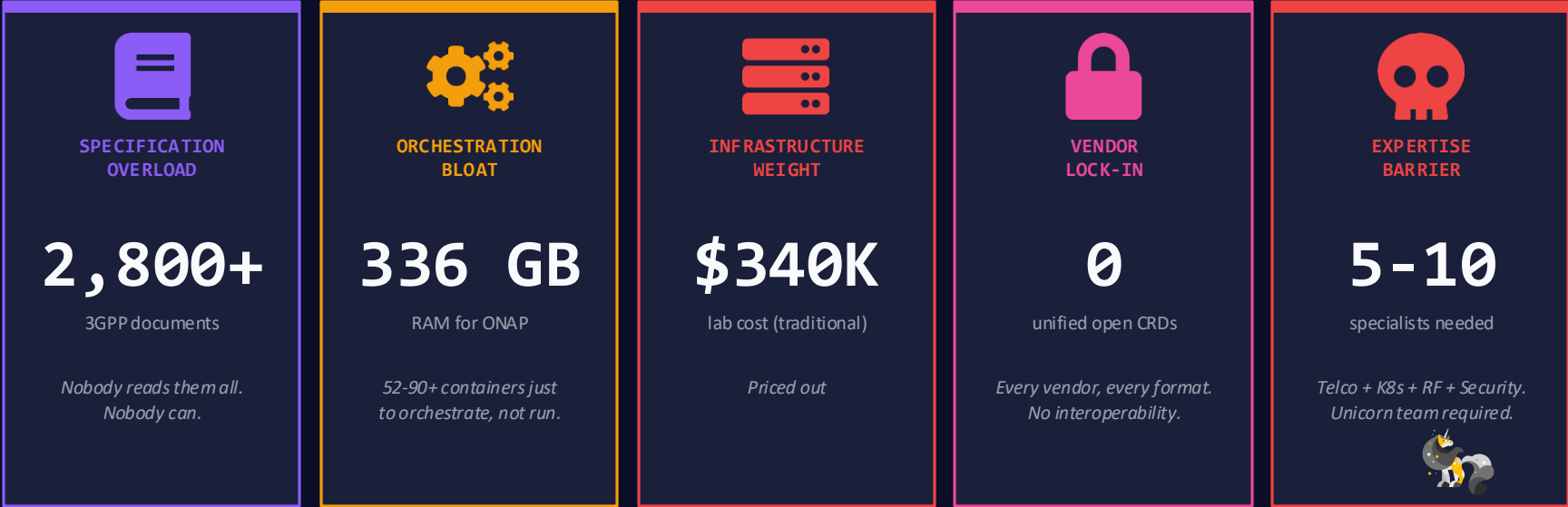
2019-2025: "Can Kubernetes handle telecom?"

2026: "YES, but not the way we use to do it"

Earlier this week, Joel Studler and Ashan Senevirathne explained the challenges of running a 5G core on Kubernetes. Today, I'm following up with a path toward the solution

THE DISEASE

Why 3GPP Private Cellular Networks Are Failing to Scale



Diagnosis: The patient is overweight, over-engineered, and over-budget. The industry built cathedrals when it needed clinics.

THE DISEASE

Why 3GPP Private Cellular Networks Are Failing to Scale



SPECIFICATION
OVERLOAD

2,800+

3GPP documents

*Nobody reads them all.
Nobody can.*

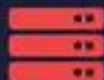


ORCHESTRATION
BLOAT

336 GB

RAM for DNMP

*52-88k containers just
to orchestrate, not run.*



INFRASTRUCTURE
WEIGHT

\$340K

lab cost (traditional)

*Priced out: universities,
startups, rural networks.*



VENDOR
LOCK-IN

0

unified open CNDs

*Every vendor, every format.
No interoperability.*



EXPERTISE
BARRIER

5-10

specialists needed

*Teleco + R&D + AI + Security.
Unknown team required.*



Diagnosis: The patient is overweight, over-engineered, and over-budget. The industry built cathedrals when it needed clinics.

Your own cellular network is one conversation away.

THE PRESCRIPTION

One CRD to describe, deploy, and heal a cellular network



PRESCRIPTION

CellularNetwork CRD

One YAML for each network

3GPP telco expert

LLM + RAG + obsidian vault

PC + SDR + K8s

\$3,550 total cost

Open + GitOps

Flux reconciliation

AI + MCP

Natural language ops

Rx

THE PRESCRIPTION

One CRD to describe, deploy, and heal a cellular network

✗ SYMPTOM

2,800+ specs

Unreadable complexity

336 GB ONAP

Orchestration bloat

\$340K lab

Unaffordable entry

Vendor lock-in

No unified model

Unicom teams

5-10 specialists



PRESCRIPTION

CellularNetwork CRD

One YAML, all 3GPP

3GPP telco expert

UEM + RAN + core/nw multi

PC + SDR + K9s

\$3,500 total cost

Open + GitOps

Flux reconciliation

AI + MCP

Natural language ops



TREATMENT

kubectl apply

5 min deployment

ngintelco install

AI control plane cellular network platform / orchestrator

Open source

Use open source software but can integrate proprietary software

Ngintelco 3gpp telco-expert

3GPP expert chatbot

Self-healing L5

3-tier AI orchestrator

Your own cellular network is one conversation away.

THE TREATMENT

One CRD. Every CNCF tool mapped.

CellularNetwork CRD + AI
apiVersion: ngintelco.io/v1alpha1

 **The Brain**
spec.core.controlPlane
AMF · SMF · NRF · AUSF · UDM · PCF

 **The Heart**
spec.core.userPlane
UPF — pumps user data through the network

 **The Lungs**
spec.ran.cu / .du / .ru
CU · DU · RU — breathes RF into the network

 **The Nervous System**
spec.lifecycle.day2
Prometheus · Grafana · Loki · Beyla (eBPF)

 **The Immune System**
spec.lifecycle.dayN
Keptn · Go rules · Nemotron LTM · Claude — self-healing

DAY 0 **Design & Provision**
Build · Operator L1: Install
Flux · Harbor/GitHub/Dockerhub
GitOps repo, container registry

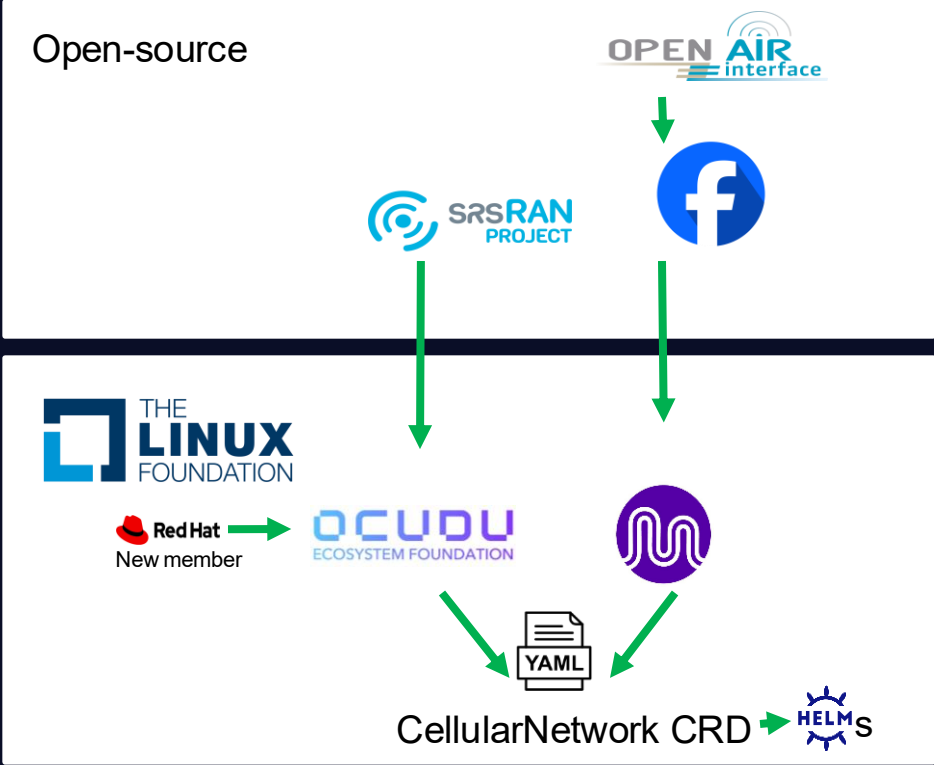
DAY 1 **Deploy & Activate**
Operate · Operator L2: Upgrade
Helm · cert-manager · NGINX Ingress/ API Gateway
Rolling deployment, TLS, service exposure, auto-rollback

DAY 2 **Observe & Optimize**
Scale & Improve · Operator L4: Deep Insights
Prometheus · Grafana · Loki · Tempo · Beyla
Metrics, dashboards, logging, tracing, eBPF auto-instrumentation

DAY N **Self-Heal & Evolve**
Adapt · Operator L5: Auto Pilot
Keptn pre and post deployments task - Go Rule Engine · Nemotron LTM 30B · other LLM API

THE TREATMENT

One CRD. Every CNCF tool mapped.



There are more
Vendor-Neutral
solutions to use



The srsRAN Project codebase has been renamed to OCUDU, and the srsRAN Project is now deprecated.
<https://www.linuxfoundation.org/press/linux-foundation-announces-ocudu-ecosystem-foundation-to-accelerate-open-source-ai-ran-innovation>

Universal 3GPP Cellular Network Blueprint



apiVersion: ngntelco.beanter.io/v1alpha1

kind: CellularNetwork

metadata:

name: enterprise-5g-sa

spec:

— Network Generation _____

generation: "5g"

— Core Network _____

core:

software: "magma" → HelmChart1

...

— RAN _____

ran:

software: "ocudu" → HelmChart1

Universal 3GPP Cellular Network Blueprint



Today:
5G SA
LTE

Future:
NB-IoT
5G Broadcast
LTE-M
NTN

...

one CRD, any network type, any vendor



THE CONVERGENCE

"The whole is greater than the sum of its parts." — Aristotle

ONE ALONE



AI



Cloud



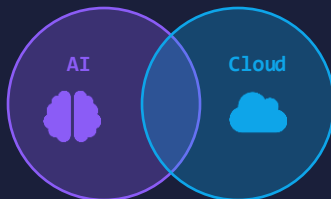
Telco

- ✗ AI alone: hallucinates 3GPP configs
- ✗ Cloud alone: no telecom domain logic
- ✗ Telco alone: manual, vendor-locked

Isolated. Incomplete.



TWO IS BETTER



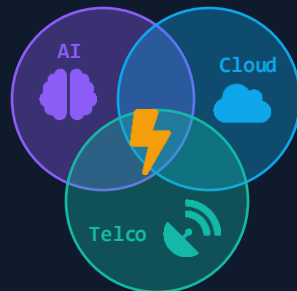
AI + Cloud

- ✓ Nephio: K8s + Telco intent (no AI)
- ✓ kagent: K8s + AI agents (no telco)
- ✓ LLM + 3GPP: generates configs (no K8s)

Better. But incomplete.



THREE CHANGES EVERYTHING



ngntelco

- 🚀 Deploy 4G/5G in natural language
- 🚀 Self-heal with 3GPP domain defined by the user + AI
- 🚀 Less MB. less pods. Less \$.
- 🚀 EU CRA compliant by design

AI-native private cellular network platform / orchestrator ngntelco

```
./ngntelco
I
NGNTELCO

Private 4G/5G Network Management Shell
We resolve the network cellular chaos
v0.1.0

Developed with ❤ by Luis Felipe Ariza Vesga
lfarizav@gmail.com

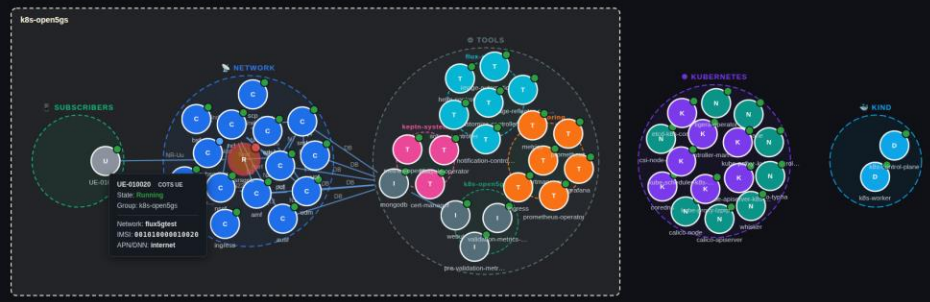
Kubecon + CloudNativeCon 2026 Edition
Amsterdam, The Netherlands
Type help for available commands or exit to quit.

Available Ngntelco Commands:

3gpp      3GPP telecommunications expert commands
clic      CLI-in-Container
completion Generate the autocompletion script for the specified shell
diag      Perform diagnostics
disrupt   Create chaos
doctor    Run health diagnostics across all infrastructure layers
extract   Extract logs, configuration, PCAPs, or infrastructure details
graph     Real-time pod/container graph viewer (Obsidian-style)
install   Install 4G/5G, 5G broadcast, NB-IoT, NTN, LTE-M networks
license   License management commands
lifecycle Manage 5G NF lifecycle checks and heartbeats
list      List all instances of a component
list-clusters List all configured Kubernetes clusters
login     Authenticate to registry
logout    Remove authentication credentials
mcp       MCP server integration utilities
observe   Open up the observation toolbox
operator  Start the ngntelco CellularNetwork operator
power     Manage and observe the power state
remove    Remove OpenSGS deployment
rollback  Roll back a Helm release to a previous revision
slack     Manage Slack integration
slice     Manage 5G network slices
ssh       Manage and connect to SSH tunnels
status    Check deployment status
subscribers Manage subscribers
test      Perform a test to evaluate the network performance
tools     Network monitoring and diagnostics tools
tunnel    Manage SSH tunnels for remote cluster access
version   Print the version number of Ngntelco CLI
watch     Real-time pod topology TUI across 5G core NFs

Tip: Use 'command --help' for detailed information about any command
```

Network Slicing



LTE/5G SA



Docker, helm, kustomize, flux

Future: NB-IoT, LTE-M, 5G Broadcast, MBS ...

AI-native private cellular network platform / orchestrator ngntelco

localhost:9191

EURECOM EURECOM RE... LATEX France MagazinesOp... departe Article Publish... geneticalgorit... LTEbasics C_basics Torrent ETS DatabasesUna...

ngntelco graph **FLUX** **SG NR** 44 nodes 29 edges 12:45:59 PM

k8s-openSgs

SUBSCRIBERS

UE-010020

NETWORK

TOOLS

KUBERNETES

Telco-models

KYM - 3GPP Expert

how many subscribers have mongodb pod

The MongoDB pod contains "22 subscriber records" (as shown by the load-subscribers job log, which validated 22 subscribers).

```
kubectl get pods -n k8s-openSgs
```

NAME	READY	STATUS	RESTARTS	AGE
k8s-allay-b4b1f6c9d-shdp	1/1	Running	1 (12m ago)	17h
k8s-anf-5f4854c7b-mah0d	1/1	Running	1 (12m ago)	17h
k8s-austf-8c8057c3d-kmsjt	1/1	Running	1 (12m ago)	17h
k8s-beyla-5ndor	1/1	Running	1 (12m ago)	17h
k8s-beyla-vlqzb	1/1	Running	1 (12m ago)	17h
k8s-bstf-80694d5c0-glcrn	1/1	Running	1 (12m ago)	17h
k8s-cloudflare-88778f7c6-pp56l	1/1	Running	1 (12m ago)	17h
k8s-lifecycle-9d3d59727-jlqjg	1/1	Running	1 (12m ago)	17h
k8s-load-subscribers-20260317-234844-fjxf7	0/1	Completed	0	17h
k8s-mongobd-56888c977-ph7z4	1/1	Running	1 (12m ago)	17h
k8s-mrf-687768f8d-wetf9	1/1	Running	1 (12m ago)	17h
k8s-nssf-8c459774-hps8d	1/1	Running	1 (12m ago)	17h
k8s-pcf-7d0d597c7-1jvvt	1/1	Running	1 (12m ago)	17h
k8s-scp-5f5d0b54b-42n67	1/1	Running	1 (12m ago)	17h
k8s-smf-56c768507-kttwc	1/1	Running	1 (12m ago)	17h

```
kubectl logs k8s-load-subscribers-20260317-234844-fjxf7 -n k8s-openSgs
```

OpenSgs Subscriber Data Validator

Validating: /config/subscribers.json

Found 22 subscriber(s) to validate

Link about your network... (Enter to send, Shift+Enter for newline)

Role: **RAN** **Core NF** **Infra** **Tools** **COTS UE** **Sim UE** **K8s CP** **CN** **Kind Node**

Tools ns: **Monitoring** **Jaeger** **Keptn** **Flux** **cert-manager**

State dot: **Running** **Pending** **Restarting** **Error** **Completed** **Unknown**

RAN srsrnb-FLUX-rn7-20 MHz - PLMN 00101 - TAC 1

Core observability lifecycle-controller lifecycle-remediation anf auct bsd ingress load-subscribers nrf nssf pcf scp smf udm ud- upf nbf-k8s-5g-core-heartbeat local-provisioner

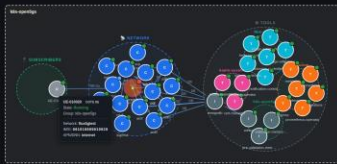
Infra mongodb webui **FLUX** helm-controller image-automation-controller image-reflector-controller customize-controller notification-controller source-controller

Deutsche Telekom

Google

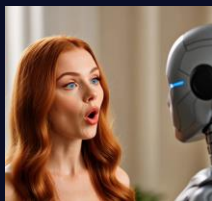
MINDR (QoS)

Ngntelco CLI orchestrates everything. AI assists through MCP but never takes the wheel



DEMO 1

**4G/5G
FLUX**



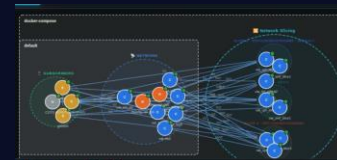
DEMO 2

**3GPP
TELCO-
EXPERT**

Luis Ariza

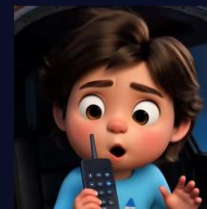


Note: No live transmission will occur, in
adherence with Authority for Digital Infrastructure
(RDI) regulatory requirements



DEMO 3

**NETWORK
SLICING (5G)**



DEMO 4

**NATURAL
LANGUAGE
5G**

DEMO 1

4G/5G NETWORK DEPLOYMENT USING KIND/HELM/K8S/FLUX/KEPTN/GRAFANA

The screenshot displays a terminal window with the following content:

```
Every 2.0s: flux get all -A.
```

NAMESPACE	NAME	REVISION	SUSPENDED	READY	MESSAGE
flux-system	gitrepository/flux-system	5q9hsl:4236500b	False	True	stored artifact
k8s-openfg	gitrepository/k8s-openfg-deploy	5q9hsl:9719003d	False	True	stored artifact
keptn-system	gitrepository/keptn-deploy	5q9hsl:44ae73bc	False	True	stored artifact
monitoring	gitrepository/monitoring-deploy	5q9hsl:789ae5ab	False	True	stored artifact

NAMESPACE	NAME	REVISION	SUSPENDED	READY	MESSAGE
k8s-openfg	helmchart/k8s-openfg-k8s	1.2.319+0715805452ab	False	True	packaged 'ke
keptn-system	helmchart/keptn-system-keptn	0.11.0	False	True	packaged 'ke
keptn-system	helmchart/keptn-system-keptn-tasks	0.1.10+44ae73bc7f	False	True	packaged 'ke
monitoring	helmchart/monitoring-prom	70.3.3	False	True	packaged 'ke

NAMESPACE	NAME	IMAGE	TAG	READY	MESSAGE
k8s-openfg	imagepolicy/k8s-openfg-policy	1/arizan/k8s-openfg	helm-sm-data-f379a20f-1755078925	True	
flux-system	flux-system	2026-03-25T19:01:04+01:00	False	True	succ

NAMESPACE	NAME	LAST RUN	SUSPENDED	READY	MESSAGE
k8s-openfg	imagesupdateautomation/k8s-openfg-all	2026-03-25T19:00:21+01:00	False	True	repo

NAMESPACE	NAME	REVISION	SUSPENDED	READY	MESSAGE
k8s-openfg	helmrelease/k8s-openfg	1.2.319+0715805452ab	False	True	helm install succeeded for
keptn-system	helmrelease/keptn	0.11.0	False	True	helm upgrade succeeded for
keptn-system	helmrelease/keptn-tasks	0.1.10+44ae73bc7f	False	True	helm upgrade succeeded for
monitoring	helmrelease/prom	70.3.3	False	True	helm upgrade succeeded for

On the right side of the terminal, there is a mobile application interface showing a grid of icons for various services like Security, Data, Analytics, and others.

Why are 3GPP standards too complex?

— Short answer. It is not complex anymore!....Wait let me reconfigure my learning screws



DEMO 2

3GPP TELCO EXPERT CHATBOT

```
Markus@Markus: ~/3GPP-Expert-Chatbot
- Regulatory bodies with diverse regional requirements
- Service providers with different business models

**2. Backward Compatibility**
- Must maintain interoperability with legacy 2G/3G/4G networks
- Gradual migration paths for existing infrastructure
- Support for multiple generations simultaneously

**3. Technical Scope**
- Radio access technologies (RAN)
- Core network architectures
- Security frameworks
- Quality of Service (QoS) mechanisms
- Mobility management
- Service enablement platforms

**4. Future-proofing**
- Anticipating unknown use cases
- Flexible architecture for emerging services
- Extensible protocols and interfaces

**5. Standards Process**
- Consensus-driven development across competing companies
- Multiple working groups with overlapping responsibilities
- Iterative refinement across releases

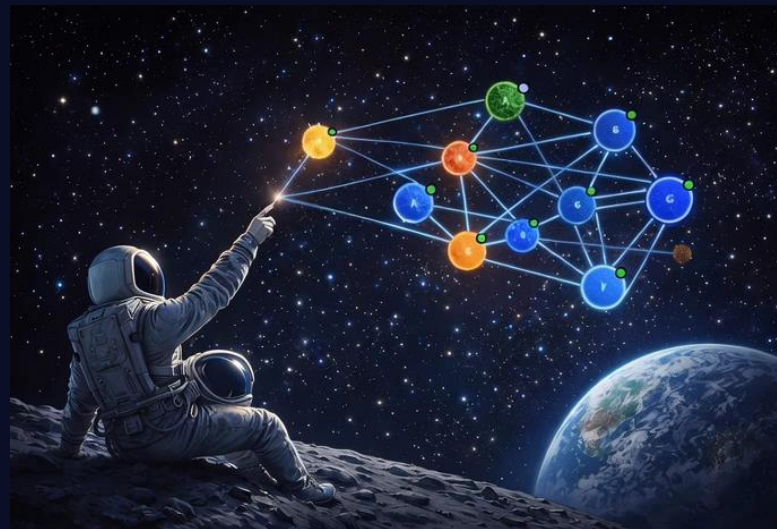
WW 3G NSA vs. 4G Networks

I believe you meant **3G NSA (Non-Standalone)** vs **4G networks**. Let me clarify the key differences:

WWW 4G LTE Network Architecture
- **Core Network** EPC (Evolved Packet Core)
  - MME (Mobility Management Entity)
  - SGMW (Serving/Gateway)
  - HSS (Home Subscriber Server)
  - PCRF (Policy Control and Charging Rules Function)
- **Radio Access** E-UTRAN
  - eNodeB (Evolved NodeB)
```

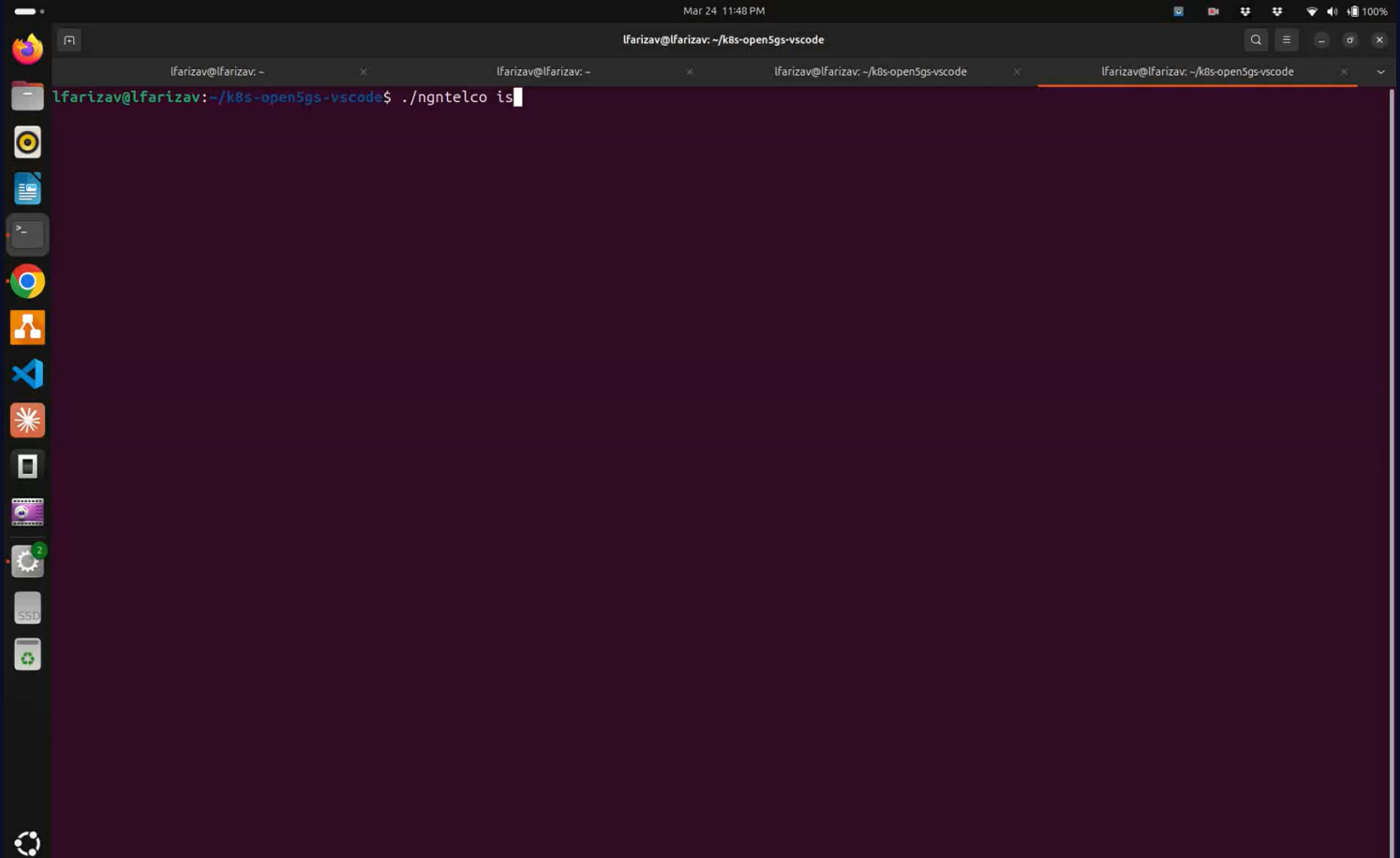
Network Slicing

— It is a technology that divides a single physical 5G network in multiple, isolated virtual slices (Different SMF, UPF, and NRF for each slice)



DEMO 3

NETWORK SLICING



“

*Daddy, I deployed a
5G network today!*

— What if I told you that the next person to
deploy a cellular network won't have a PhD?
They'll just ... ask



DEMO 4

NATURAL LANGUAGE (MCPs + ngntelco CLI)

The screenshot displays a terminal window with a dark theme. On the left, a sidebar shows various application icons. The main terminal area is split into two panes. The left pane shows the output of the command `kubectl get pods -A`, listing pods across different namespaces. The right pane shows a Claude AI chat interface with a greeting and a prompt input field.

Terminal Output (Left Pane):

```
Every 2.0s: kubectl get pods -A

NAMESPACE      NAME
calico-system   calico-apiserver
calico-system   calico-apiserver
calico-system   calico-kube-controllers
calico-system   calico-node
calico-system   calico-node
calico-system   calico-typing
calico-system   csi-node-driver
calico-system   csi-node-driver
goldmane-5f     whisker-69%
kube-system     coredns-7d
kube-system     coredns-7d
kube-system     etcd-ngntelco
kube-system     kube-apiserver
kube-system     kube-controller-manager
kube-system     kube-proxy
kube-system     kube-proxy
kube-system     kube-scheduler
local-path-storage
tigera-operator
```

Claude AI Chat (Right Pane):

Evening, Luis

How can I help you today?

Opus 4.0 Extended

Buttons: Write, Learn, Code, Life stuff, From Drive



*From a 5-year-old to 120 years old who can deploy
a 5G network... Your own cellular network is one conversation away.*

*to engineers freed from deploying cellular networks
to create real value extending code...*

*to telcos that own their future app ecosystem
instead of begging for fair share.*

That's the prescription.

From Chaos to Control.

THANK YOU



Luis Felipe Ariza Vesga • Beanter
Kubestronaut – Telco - AI

Linktree: <https://linktr.ee/lfarizav>

More demos?: I'm in the hallway with my laptop ☕

Your own cellular network is one conversation away.